Towards Vision-Spatiotemporal Fusion in Traffic Forecasting: A Survey on Cross-Modal Alignment

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Abstract

Traffic forecasting is evolving, with world models emerging as a powerful framework. By integrating real-time visuals with legacy signals, these models achieve situational awareness that transcends historical patterns. However, constructing such models presents dual challenges: a semantic misalignment between pixels and traffic concepts, and a geometric misalignment requiring spatial intelligence to register 2D inputs into 3D. Although most multimodal surveys focus on fusion architectures, this survey reframes the field of vision-spatiotemporal fusion via the unique lens of cross-modal alignment, which directly addresses the semantic and geometric failure modes limiting forecasting reliabilities. First, we organize existing methods by a taxonomy based on alignment granularity, categorizing them into three paradigms: feature-level, semantic-level, and task-level. This reveals their progression from low-level feature manipulation to high-level architectural integration. Second, we synthesize representative techniques within each paradigm, further surfacing intrinsic geometric challenges such as cross-view association and spatial mapping. Third, we examine current datasets and benchmarking practices, highlighting their deficiencies in measuring alignment quality. Finally, we discuss key future directions, including spatial intelligence for robust perception and holistic traffic world models. By providing this unified analytical framework, the survey aims to establish a new reference point for developing robust and explainable forecasting systems.

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1 Introduction

Traffic forecasting is advancing by combining live camera feeds with conventional data, shifting away from models that depend only on past patterns [Li *et al.*, 2025c]. This shift is driven by the growing networks of sensors in cities. Our societies clearly need systems that can read standard traffic metrics while also understanding the full scene provided by widespread cameras and visual sensors [Palladin *et al.*, 2025]. Creating a predictive world model for future transport therefore means blending these visual streams with movement and time-based signals such as traffic flow graphs and trajectories. The aim is to produce forecasts that are more precise, dependable, and easier to interpret [Ding *et al.*, 2025].

Merging these data streams faces a challenge deeper than just handling different formats. The core issue is a mismatch in how information is represented. Cameras deliver raw pixels, shaped by changing angles, light, and obstructions, all following optical and geometric rules [Jain *et al.*, 2025]. Forecasting models, however, work with organized structures such as graphs or time series that summarize overall traffic dynamics. This fundamental divide creates two main problems. One is a semantic gap: connecting raw visual patterns to meaningful traffic ideas. The other is a geometric gap: fitting 2D visual data into a coherent 3D world model. Common fusion techniques, such as simply joining features, often cannot bridge these gaps. As a result, model performance tends to stagnate, and adapting to new scenarios remains difficult [Dziadzio *et al.*, 2025; Doveh *et al.*, 2024].

Investigations into these issues are dispersed among distinct research communities, operating without a unified framework. For instance, substantial effort in computer vision is devoted to parsing traffic camera footage [Peddi *et al.*, 2024]. Concurrently, the spatiotemporal forecasting field advances via architectures employing graph neural networks (GNNs) and Transformers [Ke *et al.*, 2023]. Separately, some studies recast time series as visual representations for convo-

lutional neural network (CNN) processing [Lv *et al.*, 2025]. These related endeavors nonetheless leave the primary obstacle unresolved: coherently integrating real-time visual scenes with the structured models used for traffic prediction. Such fragmentation underscores the necessity for a consolidated viewpoint on aligning visual and spatiotemporal data.

Prevailing surveys in multimodal traffic forecasting commonly classify approaches by fusion strategy or timing, frequently sidestepping the essential question of meaningful cross-modal alignment. This survey proposes a distinct focus. Its central premise is that alignment granularity, the representational level at which correspondences are enforced, constitutes the fundamental challenge. Categorizing prior work via this lens reveals an evolution from superficial correlation to profound integration, yielding a pragmatic framework for assessing methodological efficacy. We present a systematic analysis of cross-modal alignment for vision-centric traffic forecasting[1]. Aiming to facilitate the development of spatially-intelligent traffic world models, this survey makes three principal contributions:

- **A taxonomy built around alignment granularity charts the route toward predictive world models.** Our framework organizes existing methods according to where they perform alignment: on features, semantics, or the task itself. This scheme makes it easier to examine and compare different lines of research.
- **A dedicated review of alignment techniques considers geometric challenges as the basis for spatial intelligence.** This survey moves beyond basic format conversion. It collects methods that fit live visual scenes into structured forecasting models and also separately dissects core geometric issues such as linking across views and spatial mapping.
- **An integrated analysis forms the groundwork for robust, spatially-aware world models.** We evaluate current datasets, benchmarks, and evaluation practices, identifying their shortcomings. From this, we recommend steps to develop more unified and effective traffic forecasting systems.

2 Preliminaries and Taxonomy

Vision-spatiotemporal fusion combines visual data with structured traffic signals. The main aim is to improve traffic prediction and situational understanding, helping to build a complete traffic world model. Here, $\mathcal{V}$ represents the visual modality, for example, video from urban cameras, and $\mathcal{S}$ represents spatiotemporal signals, such as traffic flow graphs or vehicle trajectories. A forecasting model learns a function $\mathcal{F}$ to produce predictions $\hat{\mathbf{y}}$ from both types of input:

$$\hat{\mathbf{y}} = \mathcal{F}(\mathbf{V}, \mathbf{S}; \mathbf{\Theta}), \quad \mathbf{V} \in \mathcal{V},\ \mathbf{S} \in \mathcal{S}, \tag{1}$$

where $\mathbf{\Theta}$ contains all model parameters. One main difficulty comes from the fact that visual and spatiotemporal data are very different in both meaning and structure. Bridging the semantic and geometric differences requires alignment, which is not straightforward in real urban traffic.

[1]https://github.com/Anna042023/Vision-Spatiotemporal-Fusion-in-TF

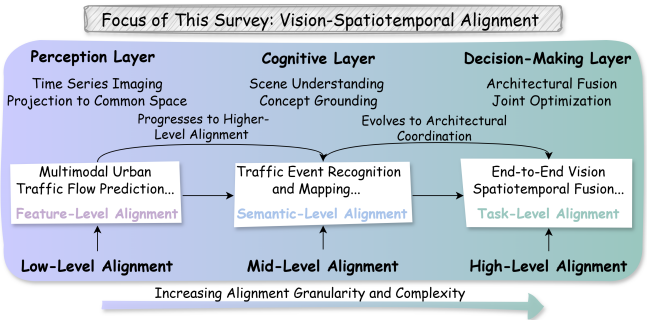

Figure 1: Levels of alignment for vision and spatiotemporal data. Methods are categorized as feature-level, semantic-level, or task-level alignment. The highlighted area indicates the main focus of this survey.

To organize the various approaches, we classify methods by the level at which alignment is carried out. This taxonomy looks at three levels: feature, semantic, and task. Figure 1 shows this range, from methods that adjust low-level features to methods that coordinate full model modules. The figure also reflects a trend in practice: vision is increasingly treated as an integral part of forecasting, rather than just a source of engineered features.

At the feature level, visual and spatiotemporal inputs are mapped into a common latent space $\mathcal{Z}$ [Liu *et al.*, 2025]. The model learns $f_v : \mathcal{V} \to \mathcal{Z}$ and $f_s : \mathcal{S} \to \mathcal{Z}$ so that the resulting representations are comparable. Approaches either transform one modality to resemble the other or embed both into a shared space. This level helps establish initial associations but often does not capture high-level traffic semantics or the geometric layout.

Semantic-level alignment links interpretable concepts across modalities [Shaw *et al.*, 2025]. Let $\mathcal{C} = \{c_1, \ldots, c_K\}$ denote traffic concepts extracted from images via $g_v : \mathcal{V} \to \mathcal{C}$, and $\mathcal{D} = \{d_1, \ldots, d_L\}$ denote states in the spatiotemporal model. A mapping $h : \mathcal{C} \to \mathcal{D}$ connects visual concepts to model states. This approach addresses the semantic gap, but it depends on accurate concept detection and proper geometric fitting.

Task-level alignment works differently. It coordinates dedicated modules for visual and spatiotemporal data, $\mathcal{M}_v$ and $\mathcal{M}_s$, in an end-to-end system [Khan and Gauch, 2025]. Their outputs are combined using:

$$\hat{\mathbf{y}} = \mathcal{F}_{\text{fuse}}(\mathcal{M}_v(\mathbf{V}), \mathcal{M}_s(\mathbf{S}); \mathbf{\Theta}), \tag{2}$$

with all parameters $\mathbf{\Theta}$ learned jointly. Here, alignment emerges automatically during training, guided only by the final prediction task. Key research questions include how to make the modules interact effectively, how to avoid dominance by one modality, and how to maintain model interpretability.

The main difference between semantic and task-level alignment is the use of intermediate concepts. Semantic alignment explicitly defines interpretable traffic concepts,

while task-level alignment integrates coordination directly into end-to-end learning. This distinction provides a straightforward way to classify the methods discussed in the following sections.

3 Vision-Spatiotemporal Alignment in Traffic Forecasting

Based on the alignment taxonomy presented earlier, this section analyzes specific methodologies for cross-modal alignment within traffic forecasting. We examine how current techniques operate at each granularity level, detailing their architectural design for traffic data and the core challenges of aligning dynamic visual inputs with structured traffic representations. The progression from feature-level to semantic-level and task-level alignment marks a clear evolution in the field, moving toward more deeply integrated systems capable of handling complex urban environments.

3.1 Feature-Level Alignment

Feature-level alignment methods aim to integrate visual observations $\mathcal{V}$ with spatiotemporal traffic data $\mathcal{S}$ by projecting them into a shared feature space $\mathcal{Z}$. These techniques primarily rely on statistical objectives that encourage similarity between features, without explicitly incorporating traffic semantics [Gong *et al.*, 2022]. This paradigm provides a foundational, though often abstract, mechanism for merging heterogeneous data streams in multimodal forecasting.

A dominant strategy involves converting spatiotemporal traffic data into visual representations. Traffic time series $\mathbf{S} = \{s_t\}_{t=1}^T$ from roadway sensors are encoded into 2D image-like formats $\mathbf{I}_s$ using transformations such as heatmaps, Gramian Angular Fields (GAF), or Recurrence Plots (RP) [Ni *et al.*, 2025]. The GAF transformation, for instance, maps a normalized traffic time series $\tilde{\mathbf{s}} = \{\tilde{s}_t\}_{t=1}^T$ to a matrix $\mathbf{G}$ with elements defined as:

$$\mathbf{G}_{ij} = \cos(\phi_i + \phi_j), \quad \phi_i = \arccos(\tilde{s}_i),\ \tilde{s}_i \in [-1, 1], \quad (3)$$

which captures temporal correlations within the signal. This conversion enables the application of pre-trained convolutional neural networks, effectively framing traffic forecasting as an image-to-image translation problem [Ferreira *et al.*, 2025]. VisionTS [Chen *et al.*, 2025] exemplifies this approach, transforming traffic flow and speed series into heatmaps to achieve strong zero-shot forecasting performance that rivals specialized models with minimal fine-tuning. A significant limitation, however, is the decoupling of the imaged traffic signal from concurrent visual streams $\mathbf{V}$ (e.g., live camera feeds). This ignores real-time semantic and geometric correspondences between the visual scene and the measured traffic states, reducing the task to signal processing rather than contextual reasoning, a critical shortcoming in dynamic urban settings where visual context directly influences traffic conditions [Zhang *et al.*, 2023].

An alternative strategy employs implicit encoding via dual-branch encoders that learn separate mappings $f_v : \mathcal{V} \to \mathcal{Z}$ and $f_s : \mathcal{S} \to \mathcal{Z}$. The alignment of their outputs, $\mathbf{z}_v = f_v(\mathbf{V})$ and $\mathbf{z}_s = f_s(\mathbf{S})$, is often driven by auxiliary contrastive learning objectives:

$$\mathcal{L}_{\text{cont}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_v, \mathbf{z}_s)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(\mathbf{z}_v^{(j)}, \mathbf{z}_s^{(i)})/\tau)}, \quad (4)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter [Huang *et al.*, 2024]. Although more flexible than explicit transformation, this paradigm frequently struggles with the inherent heterogeneity between CNN-processed images and GNN-processed traffic graphs. Minimizing feature distance alone may fail to establish semantically meaningful correspondences essential for traffic dynamics, such as associating a specific visual event (e.g., an accident) with its impact on network-wide flow.

The principal critique of feature-level alignment centers on its abstraction. By operating on low-level features, these methods obscure the high-level traffic semantics necessary for explainable forecasts. They also typically lack explicit mechanisms to resolve the geometric discrepancy between 2D image coordinates and 3D traffic network geometry. Consequently, the performance of such approaches may saturate in complex scenarios where contextual understanding is as crucial as predictive accuracy [He *et al.*, 2025].

3.2 Semantic-Level Alignment

Semantic-level alignment addresses feature-level limitations by establishing interpretable links between traffic concepts from visual streams $\mathcal{V}$ and states in structured models $\mathcal{S}$. A semantic extractor $g_v : \mathcal{V} \to \mathcal{C}$ serves as the core component, transforming raw pixels into traffic semantics $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$ such as accidents or lane blockages [Nie *et al.*, 2023]. This strategy aligns more effectively with human reasoning processes.

Early forecasting approaches depend on pre-trained detectors to extract fixed attributes such as vehicle counts. Contemporary methods utilize vision-language models (VLMs) for open-vocabulary scene understanding relevant to traffic management [Lin *et al.*, 2025a]. Structured intermediate representations such as scene graphs further enhance this process by modeling relationships between traffic entities. This improves generalization in diverse conditions where predefined ontologies prove insufficient. Recent advances integrate VLMs with scene graphs to support this grounding process [Zeng *et al.*, 2025].

Extracted semantic concepts $\mathcal{C}$ require grounding to dynamic states $\mathcal{D} = \{d_1, d_2, \ldots, d_L\}$ within the spatiotemporal model, such as predicted speed on graph edges. A grounding function $h : \mathcal{C} \to \mathcal{D}$ establishes this connection through learned or heuristic mapping. For example, the concept "congestion at intersection X" (c_k) should associate with speed reductions (d_l) on corresponding road segments. This process inherently addresses both semantic and geometric gaps, typically involving spatial registration and temporal synchronization. A standard formulation minimizes a grounding loss:

$$\mathcal{L}_{\text{ground}} = \sum_{k=1}^{K} \|\mathbf{r}(c_k) - \mathbf{q}(d_{l_k})\|^2, \quad \text{where } l_k = h(c_k), \quad (5)$$

where $\mathbf{r}(c_k)$ and $\mathbf{q}(d_{l_k})$ denote concept and state representations.

Primary challenges include the inherent ambiguity and compositionality of real-world traffic semantics, the temporal evolution of concepts, and limited availability of high-quality annotations for training robust extractors [Tang *et al.*, 2024; Wu *et al.*, 2023]. Given these constraints, developing mechanisms that extend beyond supervised learning on fixed labels remains difficult. Future directions may involve self-supervised semantic extraction from unlabeled footage and frameworks for compositional representations.

3.3 Task-Level Alignment

Task-level alignment represents the deepest integration form, embedding alignment objectives within end-to-end optimization of the complete forecasting architecture $\mathcal{F}$. A fusion function $\mathcal{F}_{\text{fuse}}$ coordinates a vision component $\mathcal{M}_v(\mathbf{V})$ and a spatiotemporal component $\mathcal{M}_s(\mathbf{S})$ directly for final forecasting performance, without explicit semantic supervision [Yi *et al.*, 2024]. This enables models to autonomously discover effective cross-modal interactions.

Common architectural designs incorporate attention-based mechanisms and gated fusion modules tailored for traffic data. Cross-modal attention layers compute relevance between a visual feature $\mathbf{v}_t$ and a traffic feature $\mathbf{s}_i$:

$$e_{t,i} = \frac{\mathbf{v}_t \mathbf{W}_q (\mathbf{s}_i \mathbf{W}_k)^\top}{\sqrt{d}}, \tag{6}$$

with corresponding attention weights:

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_{j=1}^{N} \exp(e_{t,j})}, \quad \hat{\mathbf{y}}_t = \sum_{i=1}^{N} \alpha_{t,i} \cdot \mathbf{s}_i. \tag{7}$$

Gated fusion modules dynamically weight modal contributions. A gating vector $\mathbf{g} = \sigma(\mathbf{W}_v \mathbf{z}_v + \mathbf{W}_s \mathbf{z}_s + \mathbf{b})$ controls the fusion process: $\tilde{\mathbf{z}} = \mathbf{g} \odot \mathbf{z}_v + (\mathbf{1} - \mathbf{g}) \odot \mathbf{z}_s$, where $\mathbf{W}_*$ denote learnable weights, σ represents the sigmoid function, $\mathbf{b}$ is a bias vector, and $\odot$ indicates element-wise multiplication. More recent approaches explore unified transformer-based architectures that treat visual patches and traffic graph nodes as tokens for full cross-modal attention [Lee *et al.*, 2022].

The primary strength of task-level alignment its holistic, performance-driven optimization also constitutes its main weakness for trustworthy forecasting. The implicit alignment process creates opaque behavior, obscuring learned cross-modal correspondences and complicating debugging. Modal imbalance presents another critical issue. When historical traffic data dominate near-term forecasting, models may underutilize visual streams despite their value for anomaly anticipation [Dai *et al.*, 2025]. This poses risks in safety-critical applications requiring transparency. Research challenges therefore center on designing interaction mechanisms that balance expressiveness with interpretability while managing information imbalances [Wang *et al.*, 2025; Li *et al.*, 2025d].

The central challenge involves designing fusion mechanisms that balance forecasting accuracy with interpretability for practical deployment. Promising directions include visualizing cross-modal attention patterns in traffic scenarios, introducing lightweight semantic bottlenecks to guide fusion, and developing methods to actively balance modality contributions during training.

Table 1 systematically compares representative methods across these three granularity levels. This survey categorizes alignment granularity into feature-level, semantic-level, and task-level paradigms, revealing a trade-off between interpretability and holistic performance. Feature-level methods offer transparent yet abstract transformations. Semantic-level methods align interpretable concepts but face extraction challenges. Task-level methods achieve high performance via end-to-end optimization but lack transparency. The distinction between semantic-level and task-level alignment is defined by the use of explicit intermediate semantic variables. This taxonomy clarifies the progression from correlation to deep integration and provides a framework for method evaluations.

4 The Geometry Gap in Vision-Spatiotemporal Alignment

Merging visual data $\mathcal{V}$ with structured spatiotemporal traffic data $\mathcal{S}$ faces a basic challenge that goes beyond semantic differences: the geometry gap. This problem arises from a mismatch between the 2D, projective nature of images and the 3D, metric world where traffic dynamics $\mathcal{S}$ occur. Semantic-level alignment interprets visual content but usually assumes spatial correspondence already exists. In reality, establishing this correspondence is a core geometric problem vital for fusion accuracy. Figure 2 shows how the geometry gap appears via two linked challenges that define spatial intelligence needs for world models: cross-view association and spatial coordinate mapping.

4.1 Cross-View Association

Employing distributed camera networks requires linking the same physical object across multiple, often non-overlapping views. This is essential for building consistent traffic flow pictures across a city [Yang *et al.*, 2021]. Formally, with cameras $\mathcal{C}_{\text{cam}} = \{C_1, C_2, \ldots, C_N\}$ and detected objects $\mathcal{O}_i$ in view C_i, the goal is a matching function $\mathcal{M}$ that maps objects from different views to the same global entity if they are the same physical object.

Traditional methods use re-identification (ReID) models. These learn discriminative feature embeddings, often trained with a triplet loss:

$$\mathcal{L}_{\text{triplet}} = \max\big(0, \text{dist}(\mathbf{f}_a, \mathbf{f}_p) - \text{dist}(\mathbf{f}_a, \mathbf{f}_n) + \alpha\big), \tag{8}$$

where $\mathbf{f}_a$, $\mathbf{f}_p$, and $\mathbf{f}_n$ are feature embeddings for anchor, positive, and negative samples; $\text{dist}(\cdot,\cdot)$ is a distance metric; and α is a margin. Traffic scenes add tough challenges like viewpoint changes, lighting differences, and similar-looking vehicles [Kannan and Min, 2025]. Newer methods add spatial-temporal constraints, using camera layout and travel time estimates to make association more reliable [Wang *et al.*, 2024].

Another strategy first locates objects in 3D before associating them. This reduces the problem to applying distance thresholds in 3D space. It avoids appearance confusion but needs very accurate calibration and depth estimation [Jiang

Table 1: Representative Methods for Vision-Spatiotemporal Alignment in Traffic Forecasting

Method	Align.	Visual Input	S-T Input	Mechanism	Traffic Task	Key Insight
VisionTS [Chen *et al.*, 2025]	• (F)	Time-series image (GAF)	Traffic time series	Visual MAE pre-training	Zero-shot forecasting	Transforms signals to images; bridges vision pre-training to traffic forecasting.
Dual-Encoder [Gong *et al.*, 2022]	• (F)	Traffic camera image	Graph traffic state	Bi-level adversarial alignment	Multi-modal forecasting	Task-aware data and refined feature alignment for domain adaptation.
Semantic Extractor [Nie *et al.*, 2023]	◦ (S)	Traffic video	Graph node states	Pixel-to-concept parsing	Explainable event forecast	Guides reconstruction with traffic semantics (instances, events).
Scene Graph Anticipation [Peddi *et al.*, 2024]	◦ (S)	Traffic scene images	Graph-structured events	Neural ODE/SDE modeling	Scene understanding	Models continuous dynamics of traffic entity interactions.
Gated Fusion [Yi *et al.*, 2024]	▪ (T)	Visual feature sequence	Spatio-temporal graph features	Multi-dim. spatio-temporal interaction	Multi-task forecasting	Captures cross-interactions for continuous multi-task learning.
Unified ViT-GNN Transformer [Lee *et al.*, 2022]	▪ (T)	Camera image patches	Traffic graph nodes	Cross-modality attention fusion	Pedestrian detection	Explores modality-specific features for traffic perception.

Legend: Align.: Alignment Granularity (•: Feature-level (F), ◦: Semantic-level (S), ▪: Task-level (T)). S-T Input: Spatio-Temporal Input. The table summarizes core methodologies discussed in Section 3.

et al., 2024]. Hybrid methods combine appearance features with geometric cues for better results [Sun *et al.*, 2024]. Lasting issues such as occlusions and calibration drift still hurt reliability.

4.2 Spatial Coordinate Mapping

Merging visual data with traffic models requires converting 2D image coordinates to 3D network locations. This spatial coordinate mapping is key for linking visual events to specific road segments or world coordinates.

The standard solution uses explicit geometric mapping via camera calibration. For a point on the road plane with world coordinates (X, Y), its projection to image coordinates (u, v) follows the camera projection model:

$$\gamma \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}[\mathbf{R}|\mathbf{t}] \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}, \tag{9}$$

where $\mathbf{K}$ is the intrinsic matrix, $[\mathbf{R}|\mathbf{t}]$ are extrinsic parameters, and γ is a scale factor. However, keeping accurate calibration across many cameras is hard work and error-prone. It also works poorly in non-flat scenes [Kong *et al.*, 2024].

Learning-based methods are emerging to fix these issues. One approach treats mapping as regression, training networks to predict world coordinates straight from image features [Mirzaie and Rosenhahn, 2025]. Another promising direction uses neural rendering frameworks such as NeRF. These implicitly capture dense 3D geometry from multiple views [Hua and Wang, 2024]. Once trained, such models can provide accurate 3D localization, offering automatic spatial alignment that adapts to scene changes.

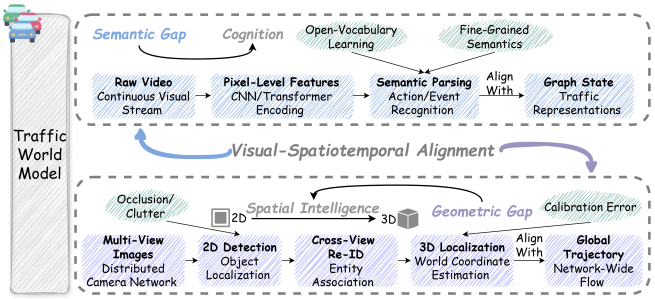

Figure 2: Dual pillars of a traffic world model: semantic and geometric gaps. The upper part illustrates the semantic gap, requiring parsing of raw video into traffic concepts aligned with spatiotemporal graph states. The lower part illustrates the geometry gap, necessitating cross-view association and 3D spatial mapping to align with global trajectories or road networks. Key technical hurdles are highlighted at each stage.

4.3 Frontiers in Geometric Learning

The field is moving toward end-to-end geometric learning. This makes spatial intelligence a core, learnable part of the world model, not a separate step. The goal is to learn geometric correspondences as latent representations useful for downstream tasks.

One frontier explores geometry-aware neural architectures. Here, cross-modal attention mechanisms use geometric priors [Shrout *et al.*, 2025]. Another area looks at integrating differentiable rendering [Zhou *et al.*, 2024]. This lets gradients flow via rendering processes to adjust geometric parameters. These approaches hint at future self-calibrating vision-spatiotemporal systems.

A major remaining challenge is how to evaluate alignment

quality. Current work mainly uses final forecasting accuracy as an indirect measure [Wu *et al.*, 2025]. Developing direct metrics, such as cross-camera association accuracy and spatial mapping error, is vital for progress. Also, datasets with high-quality geometric annotations are still rare, limiting training and testing of advanced models.

5 Resources and Evaluations

This section reviews widely adopted datasets, common evaluation protocols, and challenges encountered during real-world deployment. The objective is to depict the field's present configuration and to reveal core limitations that hinder advancement in urban traffic management.

5.1 Datasets and Benchmarks

For alignment-driven approaches to function robustly, multi-modal traffic datasets should ensure consistent synchronization across spatial and temporal axes. As outlined in Table 2, prior studies rely primarily on three real-world benchmarks along with two supplementary data forms. Accordingly, the range of attainable investigations is largely determined by the alignment properties embedded within each dataset.

Three benchmarks are particularly influential: nuScenes [Caesar *et al.*, 2020], Waymo Open Dataset [Sun *et al.*, 2020], and CityFlow [Tang *et al.*, 2019]. Each offers synchronized video, LiDAR, and map resources, together with detailed object trajectories. Forecasting models leverage these extensive spatiotemporal annotations to derive traffic graphs or sequential representations. A central alignment difficulty concerns large-scale geometric registration, specifically linking 2D image pixels to 3D map coordinates and object trajectories across multiple sensors. Although their scale enables city-level analysis, the substantial annotation cost leads to sparser and less accurate labels for fine-grained components, such as lane-level traffic flow.

Other datasets fulfill more specialized roles. CARLA [Dosovitskiy *et al.*, 2017] and comparable simulators generate synthetic data with exact ground truth, supporting analysis of rare or safety-critical scenarios despite an inherent sim-to-real discrepancy. Benchmarks constructed from time-series imagery permit evaluation of approaches that encode traffic signals as images, yet they largely avoid the more demanding semantic and geometric alignment challenges posed by real camera streams.

Datasets explicitly intended to evaluate alignment quality in forecasting remain limited. Typically, benchmarks provide raw sensor measurements together with final task labels, but exclude validated intermediate correspondences, such as cross-camera vehicle identities or precise image-to-map alignments. This lack of intermediate ground truth complicates error analysis, making it difficult to pinpoint the causes of failure when fusion-based models are applied to real traffic environments.

5.2 Critical Evaluation of Evaluation Paradigms

Traffic forecasting models are typically examined through performance on the final prediction task. For traffic speed and flow analysis, the most frequently adopted measures are Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). They are expressed as $\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|$ and $\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}$. In these formulations, $\hat{y}_i$ refers to the estimated traffic quantity, y_i corresponds to the observed ground truth, and N denotes the total number of samples. For vehicle trajectory prediction, evaluation commonly relies on Average Displacement Error (ADE) and Final Displacement Error (FDE). They are defined as $\text{ADE} = \frac{1}{T}\sum_{t=1}^{T}\|\hat{\mathbf{p}}_t - \mathbf{p}_t\|$ and $\text{FDE} = \|\hat{\mathbf{p}}_T - \mathbf{p}_T\|$, where $\hat{\mathbf{p}}_t$ and $\mathbf{p}_t$ indicate predicted and actual vehicle positions at time t, and T denotes the prediction horizon. Although these indicators are essential for comparing overall predictive capability, they capture alignment fidelity only indirectly. Strong forecasting results can emerge from predominant reliance on the spatiotemporal modality $\mathcal{S}$, while the concrete role played by visual information $\mathcal{V}$ in contextual reasoning remains ambiguous.

Placing primary emphasis on downstream accuracy therefore leaves a discernible evaluation gap. Accurately disentangling the effects of cross-modal fusion requires criteria that explicitly address alignment behavior. This requirement motivates complementary evaluation perspectives: at the feature level, through inspection of latent representation coherence; at the semantic level, through appraisal of traffic concept extraction and correspondence; and at the geometric level, through direct measurement of cross-camera association and spatial registration precision. The lack of benchmarks dedicated to these facets continues to restrict progress toward interpretable and dependable fusion designs.

In addition, existing evaluations are predominantly conducted under near-ideal conditions. They seldom challenge models with realistic alignment disturbances, including sensor desynchronization, calibration drift, or changes in scene configuration. Within current traffic forecasting benchmarks, robustness to such factors is critically important yet remains insufficiently examined [Li *et al.*, 2025e].

6 Future Directions

The analysis has identified several research gaps. This section suggests potential paths for advancing spatial intelligence and building more capable traffic world models.

6.1 Advancing Spatial Intelligence for Robust Perception

Current perception systems often fail in novel or unexpected situations [Li *et al.*, 2025a]. Enhancing their robustness necessitates a tighter integration with traffic prediction. Perception and prediction form an interdependent cycle: reliable perceptual data is fundamental for accurate prediction, while predictive context can, in turn, guide perception—for instance, by suggesting probable locations of occluded vehicles. Future architectures should therefore facilitate this joint reasoning within a unified representational space. Such a closed-loop design could support error correction: a perception error that leads to a flawed prediction might be identified and subsequently rectified through the resulting inconsistency. Such deeply integrated perception-prediction syn-

Table 2: Core multimodal datasets for vision-spatiotemporal traffic forecasting research.

Dataset	Visual Modality	Spatiotemporal Modality	Temporal Sync.	Primary Alignment Challenge
nuScenes [Caesar *et al.*, 2020]	Multi-camera, 360° views	3D object tracks, HD map elements	Yes	Large-scale geometric registration in complex intersections
Waymo Open [Sun *et al.*, 2020]	Multi-camera, LiDAR streams	Detailed vehicle trajectories, map data	Yes	Fine-grained spatial grounding for traffic participants
CityFlow [Tang *et al.*, 2019]	City-scale traffic camera network	Vehicle trajectories, ReID labels	Approximate	Cross-view association for network-wide traffic flow analysis
CARLA (Synthetic) [Dosovitskiy *et al.*, 2017]	Rendered traffic camera views	Perfect vehicle state/logs	Perfect	Sim2Real domain gap for traffic scenarios
Time-Series Benchmarks	Generated images from traffic signals	Traffic flow/speed series	N/A	Representation conversion fidelity for traffic forecasting

ergy is pivotal for developing the next generation of highly reliable traffic world models capable of adapting to complex urban dynamics.

6.2 Embodied and Interactive World Models

Embodied world models are poised to become a core component of next-generation systems. These models understand the world from the point of view of a specific agent, such as a single vehicle. They also consider that agent's possible actions. The world is not just something to observe. It changes because of the agent's own decisions and how others react to them [Lin *et al.*, 2025b]. Future research needs frameworks to model this loop. Reinforcement learning or differentiable simulators are possible tools. The key difficulty is simulating how other agents will behave in response. The model must predict how an action, such as changing lanes, influences the future behavior of nearby drivers. This moves forecasting from passive observation to interactive reasoning. It requires simulating the consequences of actions, not just predicting a fixed future. Validating these iterative interaction mechanisms in real traffic remains a major challenge.

6.3 Toward Generalizable Traffic World Models

Traffic conditions differ greatly across cities and over time. This diversity requires world models with strong generalization abilities [Vafa *et al.*, 2024]. A promising path adopts the foundation model paradigm, where a large model is pretrained on extensive and diverse traffic data. The goal is for it to learn common principles of traffic flow and visual scenes. Developing such a model faces significant hurdles. Researchers must compile varied datasets and devise self-supervised learning tasks. These tasks should enable the model to align visual and spatiotemporal data in a transferable manner for new environments. A successful general model could then serve as a foundation for many downstream applications. For instance, it can be adapted for tasks such as signal control or hazard warning with minimal additional training.

6.4 Trustworthy and Efficient World Models

For practical deployment, traffic world models must satisfy two critical requirements: trustworthiness and efficiency. Trustworthiness ensures that a model's forecasts are reliable and its decision-making process is transparent. This involves several key attributes: explainability to reveal the reasoning behind predictions, uncertainty quantification to assess the confidence level of each forecast, and robustness to maintain performance when faced with unseen scenarios, such as unusual weather or traffic incidents [Li *et al.*, 2025b]. Efficiency is equally crucial for real-world applications. Models must be computationally lightweight and fast enough to run in real-time on embedded hardware within vehicles or roadside infrastructure. A central challenge arises because the pursuit of high predictive accuracy often conflicts with these stringent efficiency needs. Therefore, the core objective is to develop models that simultaneously balance accuracy, interpretability, reliability, and speed.

7 Conclusion

This survey systematically constructs a taxonomy for vision-spatiotemporal alignment, charting its developmental path toward a predictive world model in traffic. It offers a detailed analysis of how semantic alignment establishes a cognitive layer and how spatial intelligence addresses the fundamental geometric gap, both of which are crucial for coherent world models. By critically reviewing existing datasets and evaluation paradigms, the work highlights pressing bottlenecks in the field. The proposed framework serves as a foundation for building more interpretable, reliable, and context-aware forecasting systems. Ultimately, this structured perspective is intended to guide future research toward spatially-intelligent world models that bridge modal divides for intelligent transportation.

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